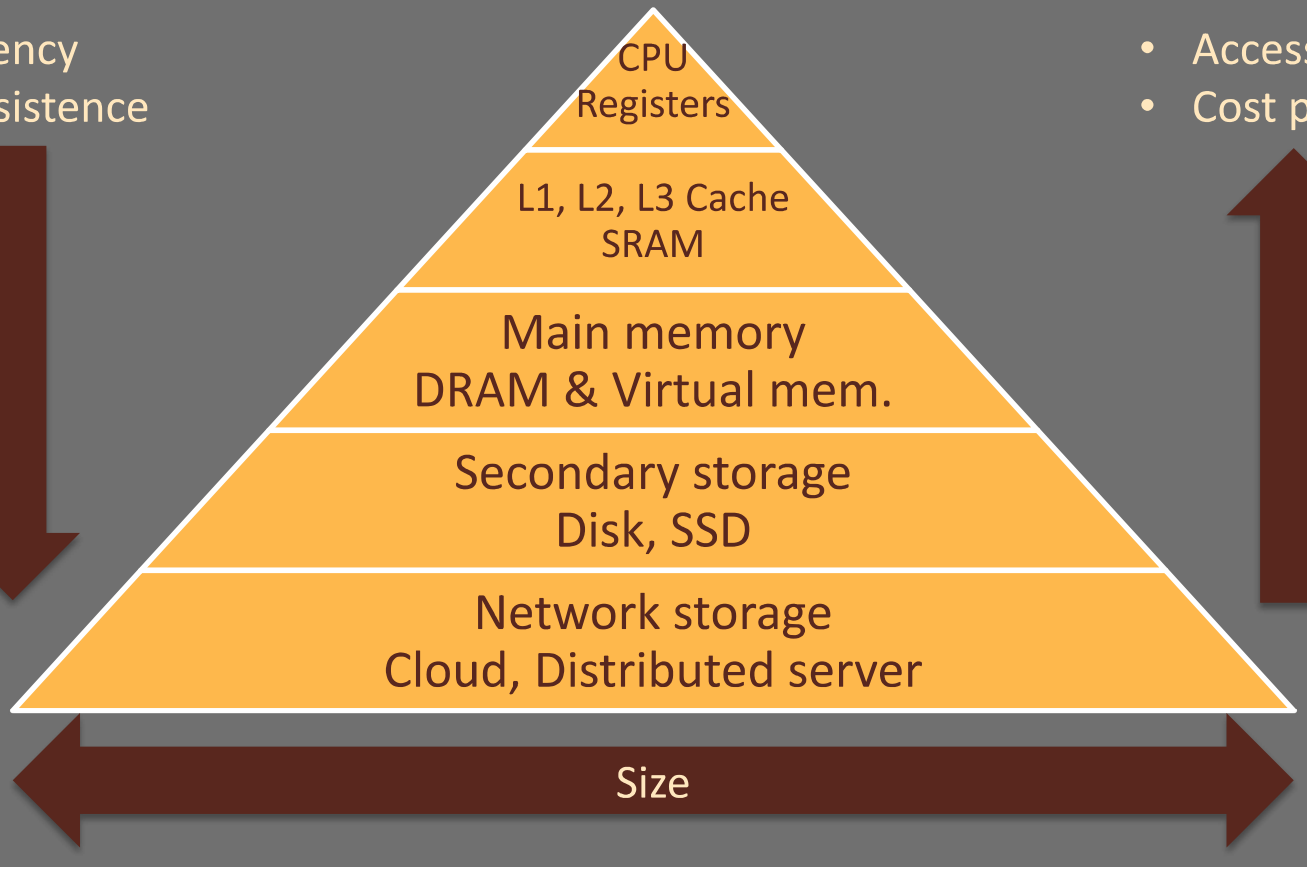
Memory Hierarchy Overview

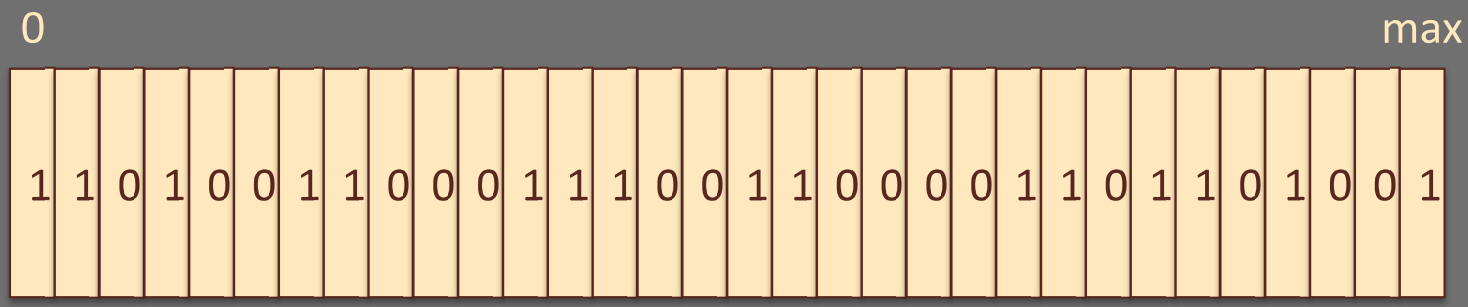
Memory Hierarchy

- Latency
- Persistence

- Access speed
- Cost per bit

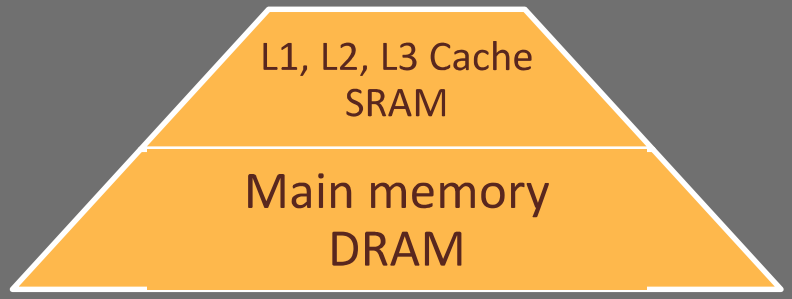


Memory map



Caching

Substitute faster cache for slower memory as much as possible.



Cache operation

Read(loc):

- If loc not in cache,
  - Load loc's memory block into cache, replacing some other block.
- Return loc's value from cache.

- Temporal locality
- Spatial locality

Main memory

Issues on

- Multiple users simultaneously.
- Multiple programs using memory simultaneously.
- Use secondary storage as backup for memory.

Main memory issues on DB server

Issue	Solution outline
Multiple DB users + other processes	Each memory block has user(s).
Insufficient memory to store everything	Memory acts as a cache for data on disk. Share memory when possible.

Memory management

VM is the standard OS solution.

But DBs usually manage their own memory.

PostgreSQL: “Shared buffers”

Best performance: 25% of physical memory.

Disk organization



Approximate latencies

1MB = 1,048,576 bytes, or the text of about 8 books

Operation	Time (microsec.)
Read 1MB sequentially from memory	250
Read 1MB sequentially from SSD	1,000
Disk seek	10,000
Read 1MB sequentially from disk	20,000
Transmit 1MB across network	5,000,000